

Database Design

1. Develop entity-relationship diagram (ERD) based on the descriptions bellow.
2. Convert the ERD to Relational model.

Descriptions of the situation

Tracking product inventories and sales of company products.

A company wants to track product inventories and sales of company products. These are some descriptions:

The company sells several products, such as clothing, cosmetic, and functional foods. The company maintains information about these products, such as product identification numbers, name, short description, weight (for shipping purposes), the category into which the product falls, a suggested price, a minimum price at which a product will be sold, manufacturer information, and order entry (OE). Products can be put in some warehouse located in some regions of the country.

Inventory information is also recorded for all products, including the warehouse where the product is available, the quantity on hand (amount in stock), short explanation in case of out of stock, and the maximum unit of the product can be put in the warehouse.

The company maintains warehouses in several locations to fulfill customer needs. Each warehouse has a warehouse identification number, name, facility description, and region.

Customer information is also tracked. Each customer has an identification number. Customer records include customer name, address, phone number, email, postal code, and notes.

The company maintains information about employees: identification number, name, the day of starting to work in the company, title (warehouse manager, administration, Sales Representative, head of a department...), salary per a month. Each employee belongs to only one department.

When a customer places an order, the company tracks the date of the order, how the order was placed (form submitted over the Internet, phone call, fax, email...), the current status of the order, payment type, address for shipping, total amount of the order, and the sales representative who helped place the order. If an order is placed over the Internet, no sales representative is recorded. In addition, to order information, the company also tracks the number of items ordered, the unit price, and the products ordered.